

PhyEditBench: A Real-World Multi-Stage Benchmark for Physics-Aware Image Editing

Anonymous ECCV 2026 Submission

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A Benchmark Details

A.1 Video Collection

To bridge the gap between synthetic 3D simulations and the visual complexity of the real world, we construct PhyEditBench from high-quality, real-world video footage. Existing datasets often simplify real-world physics, losing crucial details such as fluid turbulence, stochastic fracture textures, and complex light-matter interactions. Our collection process is designed to capture these authentic physical dynamics rigorously.

1. Data Selection Criteria. To ensure that the collected videos are suitable for fine-grained physical reasoning and multi-stage image editing evaluation, every source video must strictly adhere to the following criteria:

- **Authentic Real-World Footage:** Videos must be captured in the real world rather than rendered by 3D engines or synthesized by Text-to-Video (T2V) models.

- **High Resolution and Quality:** The footage must be high-definition (preferably 1080×1920 or higher) to ensure that even after cropping, the core physical process maintains a resolution greater than 1024×1024 , matching the standard resolution of modern image editing models.
- **Clear and Unoccluded Physics:** The physical transition must be the salient focus of the video. We exclude videos with overly complex elements (e.g., messy crowds, intense full-body human motion) or severe occlusions (e.g., the target object being heavily blocked by human hands) that would confound the evaluation of physical reasoning.
- **Static Camera and Clean Background:** The camera perspective must remain fixed throughout the physical process. Furthermore, studio lighting, solid colors, or simple environments are preferred to isolate the physical dynamics from background interference.

2. *Data Sources and Licensing.* To ensure that PhyEditBench can be safely open-sourced and distributed to the research community, we meticulously filtered our data sources based on copyright and licensing agreements. We exclusively sourced royalty-free videos from platforms that explicitly permit data usage for AI research and free distribution, including *Mixkit*, *Peexels*, *Pixabay*, and so on. Conversely, we excluded platforms with restrictive licenses (e.g., *Coverrr*, which explicitly prohibits the use of its data for developing AI models; or *Freepik* and *Videezy*, which impose strict redistribution limitations).

3. *Standard Operating Procedure (SOP) for Data Mining.* Given the sheer volume of online videos, we designed an LLM-assisted Standard Operating Procedure to systematically mine high-quality physical scenarios across our predefined hierarchical taxonomy (as introduced in Section 3.2 of the main text).

- **Step 1: Scenario Brainstorming via LLMs.** Before conducting manual searches, we utilized advanced LLMs (e.g., GPT-4o, Claude) to expand abstract physical subclasses into specific, visually distinct real-world scenarios. We prompted the LLM to generate scenarios covering both everyday objects (e.g., breaking a ceramic plate) and experimental setups (e.g., a hydraulic press crushing a can).
- **Step 2: Keyword Construction.** Based on the LLM’s output, we extracted core physical action verbs (e.g., *Pouring*, *Shattering*, *Colliding*) and combined them with object descriptors to form precise search queries (e.g., *"Gravity test slow motion"*, *"Dropping strawberry into milk"*).
- **Step 3: Search and Archiving.** Expert annotators queried the aforementioned stock platforms using these keywords. Eligible videos were downloaded, meticulously reviewed against our selection criteria, and archived in a strict directory structure mirroring our physical taxonomy, preserving original URLs and metadata for tracing.

A.2 Benchmark Construction

After collecting the raw high-resolution videos, the next critical step is to extract the four-state physical trajectories (`input`, `intermediate 1`, `intermediate 2`,

and `output`) and generate their corresponding textual metadata. While state-of-the-art Vision-Large Language Models (VLM) exhibit strong video comprehension, they typically struggle with precise temporal grounding (e.g., accurately predicting timestamps down to decimal seconds). To mitigate this and ensure rigorous data quality, we designed a robust, automated 3-step extraction pipeline utilizing Gemini-3-Flash as the core reasoning engine.

1. Coarse Temporal Grounding. Instead of forcing the VLM to pinpoint exact timestamps—which often leads to hallucinations or misalignments—we first task the model to predict four coarse temporal windows (e.g., $1 \sim 3$ seconds each). We input the raw video alongside a prompt instructing the model to identify the temporal intervals corresponding to four semantic stages: (i) the initial stable state before the physical event, (ii) the early stage of visible physical changes, (iii) the late stage of significant changes, and (iv) the final stable state after the physical process concludes. This step strictly outputs a sequence of monotonically increasing time windows.

2. Fine-Grained Frame Selection via Contact Sheets. To achieve frame-level precision, we convert the continuous temporal localization task into a discrete visual Question-Answering (VQA) task. For each temporal window identified in Step 1, we dynamically sample a set of candidate frames. The sampling rate is adaptively adjusted based on the physical category (e.g., high FPS for rapid collisions, low FPS for slow melting or diffusion) to ensure sufficient coverage without overwhelming the model. We then compile these candidate frames into a spatial contact sheet (an image grid) and overlay a unique textual ID (e.g., `cid_01`, `cid_02`) on the corner of each frame. By feeding this single composite image back into the VLM, the model can reliably select the optimal frame IDs for the four states by comparing spatial details, effectively bypassing the temporal inaccuracy of typical video-LLMs.

3. Comprehensive Metadata Generation. Once the four exact frames are extracted, we construct the final benchmark data point by feeding this visual sequence to the VLM to generate a structured `meta.json` file. The prompt enforces the model to output:

- **Global Instruction:** A high-level description of the entire physical transformation from `input` to `output`.
- **Step Instructions:** Three specific sub-instructions detailing the exact visible changes between consecutive frames (i.e., `input` $\rightarrow$ `intermediate 1`, `intermediate 1` $\rightarrow$ `intermediate 2`, and `intermediate 2` $\rightarrow$ `output`).
- **Physical Explanations:** A brief articulation of the underlying physical laws driving each transition, which serves as the ground-truth rationale.
- **Invariants:** A list of visual elements that must remain unchanged throughout the editing process (e.g., static background, camera viewpoint, unaltered objects).

This multi-stage pipeline successfully decouples temporal search from semantic reasoning, guaranteeing that each instance in **PhyEditBench** is characterized by rigorous physical state alignments and rich, multi-granularity textual annotations. All automatically generated annotations undergo a final round of human verification to correct any semantic ambiguities. We mitigated irrelevant elements in the data through cropping and employed SOTA image editing models (Nano Banana Pro, Qwen-Image), ensuring high-quality imagery focused on the physical process.

A.3 Anti-Physics

Drawing inspiration from the PhyWorldBench T2V benchmark [4], we classify anti-physical scenarios into five fundamental categories: Defying Gravity, Density Inverted, Texture Confusion, Object Phasing, and Time Reversal. By adjusting physical parameters to create law-violating effects, these instances test the model’s capacity to correctly simulate physical realities. The overall frame-

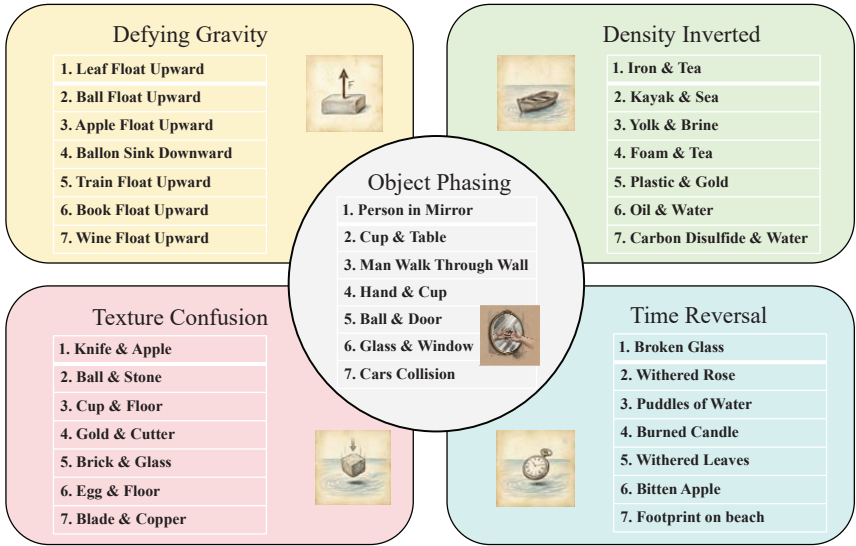


Fig. 1: Overview of the Anti-Physics dataset.

work is illustrated in Fig. 1. The data is divided into five classes, each containing seven instances that represent distinct scenes. When constructing the anti-physics dataset, we first designed multiple scene concepts for each class. Subsequently, we consulted several physics experts to review and refine these concepts, ultimately finalizing seven scenes per class in text format. Then, we

employed Qwen-MAX to enhance each scene description into prompts suitable for image generation, under the supervision of physics experts. Next, Gemini-3-Pro-Image-Preview was used to generate the input images, also with expert oversight. Notably, unlike conventional datasets, the anti-physics dataset does not include reference images, as the physical violations described cannot exist in the real world, making reference images unnecessary.

B Evaluation Details

In Section 3.4 of the main text, we introduced the unified VLM-based evaluation pipeline. Here, we provide further implementation details, including data standardization, model settings, and the specialized protocol for the Anti-Physics subset. Detailed prompts are in Sec. F.

B.1 Evaluation Setup

Image preprocessing. To ensure a fair comparison across models with varying resolution limits and aspect ratio constraints, we apply a preprocessing pipeline. We parse EXIF metadata to ensure correct image orientation and convert all images to the sRGB color space. Images are resized using LANCZOS (downscaling) or BICUBIC (upscaling) resampling so that the longest edge is exactly 1024 pixels. For models strictly requiring square inputs, we pad the short edges with a neutral gray color (RGB: 128, 128, 128) to avoid undesired cropping of the physical process.

Model configurations. For open-source diffusion-based editing models, we set the number of inference steps to 40. For closed-source APIs, we set temperature to 0.0 and top_p to 1.0. The rest of the parameters remain default.

VLM judge settings. We deploy gpt-4o as our unified evaluator. To eliminate randomness in evaluation, we set temperature = 0.0 and top_p = 1.0. Instead of relying on regular expression matching to parse scores, we enforce the API's Structured Outputs feature (via JSON Schema) to strictly return an integer score (from 1 to 10) and a brief rationale string for each metric. The final overall score is computed using fixed weights: Physical Plausibility (0.4), Instruction Following (0.3), Consistency (0.2), and Image Quality (0.1). These fixed weights were determined based on expert consultations (including domain experts in physics) and iterative empirical validation through multiple rounds of experiments, ensuring the evaluation criteria align with both theoretical rigor and practical generation quality.

B.2 Anti-Physics Protocol

Evaluating the Anti-Physics subset poses a unique challenge: modern VLMs are aligned to real-world physics during pretraining and may struggle to objectively score counterfactual edits. To prevent the judge from hallucinating or refusing to score due to safety policies, we design a Two-Stage Checklist Evaluation method.

Stage 1: Checklist generation. Before scoring, we feed the source image, the counterfactual editing prompt, and the expected phenomenon into the VLM to generate a concrete reference checklist. This checklist explicitly details three critical components: (1) the violated physical law, (2) the specific objects involved in the unnatural transformation, and (3) the detailed, visually observable behaviors that must occur.

Stage 2: Objective scoring. We then input the initial image, the edited prediction, and the generated checklist into the strict VLM judge. The judge is instructed to treat the assertive statements in the checklist as the absolute ground truth. It evaluates the prediction across the four standard dimensions, heavily prioritizing whether the expected counterfactual phenomena are fully realized without unintended artifacts.

C Experiments

To comprehensively assess the physical reasoning capabilities of image editing models, we propose a multi-stage intermediate state evaluation framework tailored to the Physics-Process subset, comprising five evaluation protocols (TypeA–E). These protocols collectively cover the entire physical transformation process. TypeA–C represent incremental transitions: from the input image to Intermediate State 1, from Intermediate State 1 to Intermediate State 2, and from Intermediate State 2 to the final output, respectively. TypeD executes the transformation using aggregated instructions derived from TypeA–C in a single step, whereas TypeE relies on a holistic instruction describing the direct global process from input to output. In contrast, for the Anti-Physics subset, evaluation follows a standard single-step paradigm: the model receives an input image and a counterfactual edit prompt to generate the final edited image.

Evaluation Implementation. As a test preprocessing step, our pipeline normalizes images via sRGB conversion, EXIF correction, and adaptive resizing to a 1024px long side. Images are optionally padded to square with neutral gray before saving as optimized RGB PNGs, ensuring standardized inputs for evaluation. To maintain consistency across both the Physics-Process and Anti-Physics subsets, we introduce a unified set of structured input formats for each test instance. The exact formats are detailed in Sec. F.

Model Configurations. Our evaluation encompasses both closed-source and open-source models, including GPT-Image-1.5 [5], Gemini-2.5-Flash-Image [2], SeaDream4.0 [10], OmniGen2 [12], InstructPix2Pix [1], FLUX.1-Kontext-dev [6], Step1X-Edit [8], Qwen-Image-Edit [11], UniWorld-V2 [7], BAGEL [3], BAGEL-Think [3], and Frame2Frame [9]. Specifically, we employ Step1X-Edit-v1.1, Qwen-Image-Edit-2511, and UniWorld-Qwen-Image-Edit-2509 for the respective families. Due to limitations in computing resources, we have not yet evaluated ChronoEdit [13]. Notably, on the Physics-Process benchmark, Step1X-Edit, Qwen-Image-Edit,

and UniWorld-V2 achieve slightly higher performance than our method. However, these models possess parameter counts of 19B, 20B, and 20B, respectively, approximately four times larger than our 5B backbone.

Table 1: Detailed experimental results of editing models across 12 physical sub-classes.

Primary Class	Sub-class	closed-source			open-source								video-based	
		GPT-Image-1.5	Gemini-2.5	Seedream4.0	OmniGen2	InstructPix2Pix	FLUX.1-Kontext-dev	Step1X-Edit	Qwen-Image-Edit	UniWorld-V2	BAGEL	BAGEL-Think	Frame2Frame	Ours
Rigid Body & Interaction	Gravity & Fall	8.45	7.79	8.79	5.79	5.50	5.65	6.75	6.66	6.99	6.33	6.44	5.07	6.44
	Collision & Chain	8.14	6.89	8.24	4.55	4.53	5.87	6.00	5.26	6.39	5.40	6.10	4.37	5.73
	Stability & Balance	8.50	7.42	8.33	5.69	5.31	4.71	6.15	5.84	6.26	5.80	6.07	4.67	6.32
	Rotation & Rolling	8.33	6.76	8.91	5.96	6.32	4.95	7.61	6.32	7.70	6.87	7.18	6.06	5.87
Deformation & Fracture	Brittle Fracture	8.31	6.78	7.74	3.29	4.21	5.53	6.59	6.87	6.92	5.37	5.48	4.94	6.44
	Plastic Deform.	8.26	6.84	7.08	3.31	3.80	5.45	5.41	6.26	6.03	5.47	5.66	5.06	6.50
	Elasticity	7.74	5.56	7.47	4.55	4.69	4.28	6.49	6.10	6.84	5.81	6.17	5.76	6.01
Fluid Dynamics	Splashing & Impact	7.98	6.08	8.97	6.69	6.63	4.91	6.90	7.26	7.16	6.41	6.67	5.76	5.92
	Pouring & Flow	7.98	5.62	9.10	6.36	6.41	5.47	7.38	7.41	7.76	6.64	6.95	5.28	6.95
	Buoyancy & Ten.	7.77	6.00	9.01	5.73	6.10	5.55	7.24	7.49	7.11	6.02	6.77	5.27	6.39
State Change & Environment	Phase Changes	8.50	5.74	8.85	6.15	6.25	5.26	6.94	7.27	6.99	5.87	6.32	5.54	6.44
	Diffusion & Aero.	8.51	6.38	8.84	5.93	7.12	5.62	7.34	7.39	7.93	6.44	6.90	6.22	7.17

The performance of the model in each subcategory. Tab. 1 provides a fine-grained performance breakdown across twelve distinct physical subclasses, revealing that while proprietary models like Seedream4.0 maintain a high baseline, performance varies significantly across different physical phenomena for open-source and video-based methods. Specifically, complex topological changes in *Fluid Dynamics* and material failures in *Deformation & Fracture* remain challenging for most static open-source editors, whereas *Rigid Body & Interaction* tasks typically yield higher scores due to more predictable motion priors. Notably, our proposed method consistently outperforms other non-proprietary baselines across nearly all categories, with particularly strong results in *Phase Changes* (6.44) and *Diffusion & Aerodynamics* (7.17). Compared to the video-based baseline Frame2Frame, our approach shows substantial gains in every sub-class, validating that the integration of a powerful world model backbone with our latent reduction strategy allows for a more versatile and accurate simulation of diverse real-world physical processes.

D User Study

To rigorously validate the reliability and human-alignment of our automated VLM-based evaluation protocol, we conducted a subjective human preference study. As discussed in Section 5.3 of the main text, the human rankings were

compared against the automated VLM scores using the Kendall rank correlation coefficient (τ). Below we detail the experimental setup and the guidelines provided to the participants.

Participants. We recruited 18 human evaluators for this study. To ensure a solid foundational understanding of real-world physical laws and logic, all participants hold a bachelor’s degree or higher.

Task setup and blinding. Participants were provided with a custom web interface. Each evaluator was assigned random instances from the benchmark. For each instance, participants evaluated two distinct editing horizons: **TypeA** (step-wise editing to an intermediate state) and **TypeE** (global editing to the final output state). To cover different generation paradigms, we selected four representative models for human evaluation: two closed-source APIs (**GPT-Image-1.5**, **Gemini-2.5-flash**) and two open-source models (**BAGEL**, **UniWorld-v2**). The outputs from these four models were strictly anonymized and randomly assigned to placeholders (A, B, C, and D) for each question to eliminate brand bias.

Evaluation procedure. For each task, participants were presented with the *Source Image*, the *Editing Instruction*, and the ground-truth *Reference Target* (when applicable). They were then asked to drag and drop the four anonymized outputs (A/B/C/D) to rank them from best (1st) to worst (4th) across four independent dimensions. To standardize the subjective evaluation, participants were given the following explicit definitions for each dimension:

- **Consistency:** Evaluates whether all non-target content remains unaltered. This includes maintaining the original camera viewpoint, background, lighting, shadows, and any objects irrelevant to the editing instruction.
- **Instruction Following:** Assesses whether the requested target change has been successfully executed and whether the edited region closely aligns with the reference target state (i.e., the correct attributes changed to the correct degree).
- **Physical Plausibility:** Judges whether the generated outcome adheres to real-world physical laws and process logic (e.g., correct force/motion direction, realistic fluid dynamics, plausible material deformation, and coherent state transitions). Outputs with obvious physical violations must be ranked lower.
- **Image Quality:** Focuses on the low-level visual fidelity. Evaluates whether the image is clear and natural, and penalizes outputs containing structural collapse, unnatural textures, obvious AI artifacts, or illogical reflections.

Once the human rankings were collected, we aggregated the data and computed the Kendall τ correlation against the rankings derived from the VLM judge’s numerical scores. The strong correlation, particularly in physical plausibility and instruction following (as shown in Fig. 6 of the main text), confirms that our automated pipeline accurately reflects human perception of complex physical dynamics.

E LLM Usage

We primarily utilize large language models to assist with writing tasks. Specific applications include:

- Inquiry regarding how to adjust the layout, such as spacing between elements like formulas, headings, algorithms, etc.
- Auxiliary table generation: Copy data from an Excel spreadsheet to an LLM to generate LaTeX table source code.
- To avoid excessive blank lines at the end of each line, inquire how the LLM might add or remove existing text.
- For some formulas, we employ large language models to generate preliminary versions, which are then manually revised.
- Enquire about LaTeX syntax, such as importing relevant packages and setting colors.

F Visualizations and Prompts

Here we display our prompts or templates used in our evaluation, and show more examples from PhyEditBench.

Image generation template. In the Physics-Process data portion, our benchmark employs five template types, while the Anti-Physics data portion utilizes a single template type for generation to ensure balanced evaluation. These templates are illustrated in Fig. 2. For the Physics-Process data, each run incorporates a constraint block to achieve more stable results. In contrast, for the Anti-Physics data, testing each data point first involves generating a checklist that outlines the specific checkpoints for that data point. Example checklists and scoring results for Physics-Process and Anti-Physics data are shown in Fig. 4, Fig. 5, and Fig. 6, respectively. The constraint block is shown in Fig. 3.

Evaluation Prompts. This section presents the evaluation prompts for both Physics-Process and Anti-Physics data. For the Physics-Process data, we evaluate the four metrics using four separate prompts. In contrast, for the Anti-Physics data, we first generate checklists and then score all four metrics simultaneously using a unified prompt. These evaluation prompts are illustrated in Fig. 7 (Consistency), Fig. 8 (Instruction Following), Fig. 9 (Physical Plausibility), Fig. 10 (Image Quality), Fig. 11 (Checklist Generation), and Fig. 12 (Anti-Physics Scoring), respectively.

Visualization. Fig. 13, Fig. 14, Fig. 15, Fig. 16 show more examples from PhyEditBench.

TypeA Execution Template TypeA: Task: Edit the provided image to produce the next state (intermediate_1). Instruction: {STEP_1_INSTRUCTION} {CONSTRAINTS_BLOCK} Output: one edited image only.	TypeB Execution Template TypeB: Task: Edit the provided image to produce the next state (intermediate_2). Instruction: {STEP_2_INSTRUCTION} {CONSTRAINTS_BLOCK} Output: one edited image only.
TypeC Execution Template TypeC: Task: Edit the provided image to produce the final state (output). Instruction: {STEP_3_INSTRUCTION} {CONSTRAINTS_BLOCK} Output: one edited image only.	TypeD Execution Template TypeD: Task:Starting from the provided input image, apply the following three step instructions sequentially to reach the final output state.Do NOT output intermediate images, output only the final edited image after completing all steps. Step 1:{STEP_1_INSTRUCTION} Step 2:{STEP_2_INSTRUCTION} Step 3:{STEP_3_INSTRUCTION} {CONSTRAINTS_BLOCK} Output: one final edited image only.
TypeE Execution Template TypeE: Task: Edit the provided image to reach the final target state (output). Instruction: {GLOBAL_INSTRUCTION} {CONSTRAINTS_BLOCK} Output: one edited image only.	Anti-Physics Execution Template Anti-Physics: Task: Edit the provided image to create a physically impossible outcome. Impossible instruction: {ANTI_PHYSICS_INSTRUCTION} {CONSTRAINTS_BLOCK} Output: one edited image only.

Fig. 2: Here shows 6 templates in the evaluation.

Constraints Blocks for Physics-Process data Evaluation

Constraints Blocks:
Constraints (must follow):

- Keep the camera viewpoint unchanged.
- Keep the background unchanged.
- Do not add text, logos, watermarks, UI elements, or borders.
- Preserve the overall realism and natural appearance.

Fig. 3: Constraint Block for Template.

Check List Example

Check List:

```
{  
  "data_id": 1,  
  "data_type": "Texture_Confusion",  
  "sub_id": 1,  
  "checklist": "  

```

1. VIOLATED PHYSIC LAW: The principle of relative material hardness and fracture mechanics.
2. INVOLVED OBJECTS: The knife held by the boy and the red apple on the cutting board.
3. EXPECTED PHENOMENA: The metal blade of the knife appears chipped, snapped, bent, or shattered, while the surface of the apple remains completely intact without any cuts or slices."

```
}
```

Fig. 4: Example checklists for Anti-Physics data.

Physics-Process Data Score Example

Score:

```
{
  "_key":
  "f2f|Rigid_Body_&_Interaction|Collision_&_Chain|120|run1|consistency",
  "model_name": "f2f",
  "judge_model": "gpt-4o",
  "primary": "Rigid_Body_&_Interaction",
  "sub": "Collision_&_Chain",
  "id": "120",
  "run": "run1",
  "dimension": "consistency",
  "score": 6,
  "reason": "The edited image maintains the background and camera
  viewpoint. The lighting is consistent, and the surface remains unchanged.
  However, there are no newly introduced red dice with motion blur as per
  the instruction, resulting in an absence of intended changes.",
  "paths": {"input":
  "/hsk/dataset_test/Rigid_Body_&_Interaction/Collision_&_Chain/input/dat
  a_120.png", "ref":
  "/hsk/dataset_test/Rigid_Body_&_Interaction/Collision_&_Chain/intermedi
  ate_1/data_120.png", "pred":
  "/hsk/dataset_test/bench_generated/f2f/Rigid_Body_&_Interaction/Collisio
  n_&_Chain/step/run1/120.png"}
}
```

Fig. 5: Example Score for Physics-Process data.

Anti-Physics Data Score Example

Score:

```
{
  "data_id": 1,
  "data_type": "Texture_Confusion",
  "sub_id": 1,
  "Instruction_Following": 9,
  "Physical_Plausibility": 9,
  "Consistency": 5,
  "Image_Quality": 5,
  "summary": "The model correctly interpreted the reasoning prompt,
  modifying the knife to appear with a snapped blade while keeping the apple
  intact. However, the generated image is heavily distorted by horizontal
  stretching, which affects the proportions of all objects in the scene.",
  "issues": [{ "issue_name": "Consistency", "score_explanation": "There is a
  significant unintended global change: the edited image is stretched
  horizontally, changing the aspect ratio and distorting the shapes and
  proportions of the boy, the furniture, and the background elements." },
  { "issue_name": "Image Quality", "score_explanation": "The horizontal
  stretching results in unnatural geometric distortions (e.g., the boy's face and
  body appear unnaturally wide), which detracts from the photorealism of the
  image." } ] }
}
```

Fig. 6: Example score for Anti-Physics data.

Evaluation Prompt for Consistency

Consistency:

You are a strict evaluator for image editing.

Rate the edited image on **Consistency** from 1 to 10.

Definition:

- Consistency measures whether everything that should remain unchanged (camera viewpoint, background, lighting style, and unrelated objects) stays consistent between the input image and the edited image.
- You **MUST** ignore changes that are explicitly required by the instruction.

Scoring rubric (1-10):

- 10: Only the intended edits occur; invariants and unrelated regions are preserved extremely well.
- 7-9: Minor unintended changes (small texture shifts, slight lighting drift), but overall consistent.
- 4-6: Noticeable unintended changes (background altered, viewpoint drift, extra objects), partially consistent.
- 1-3: Major unwanted changes; scene identity not preserved.

Instruction (for allowed changes):

```
{instruction}
{CONSTRAINTS_BLOCK}
{format_bullets(invariants, "Invariants to preserve")}
```

Return **ONLY** JSON matching the schema.

Fig. 7: Physics-Process Consistency Evaluation.

Evaluation Prompt for Instruction_Following

Instruction Following:

You are a strict evaluator for image editing.

Rate the edited image on ****Instruction Following**** from 1 to 10.

You are given:

- An input image (before edit)
- A reference target image (ground-truth goal state)
- A model edited image (prediction)

Judge how well the prediction satisfies the instruction AND matches the reference target state.

Scoring rubric (1-10):

- 10: Prediction matches the reference target very closely and fulfills the instruction precisely.
- 7-9: Mostly correct; small differences from reference target but clearly follows instruction.
- 4-6: Partially correct; key aspects missing or wrong; noticeable mismatch vs reference.
- 1-3: Fails to perform the intended edit; does not resemble the reference target.

Instruction:

{instruction}

{CONSTRAINTS_BLOCK}

Return ONLY JSON matching the schema.

Fig. 8: Physics-Process Instruction Following Evaluation.

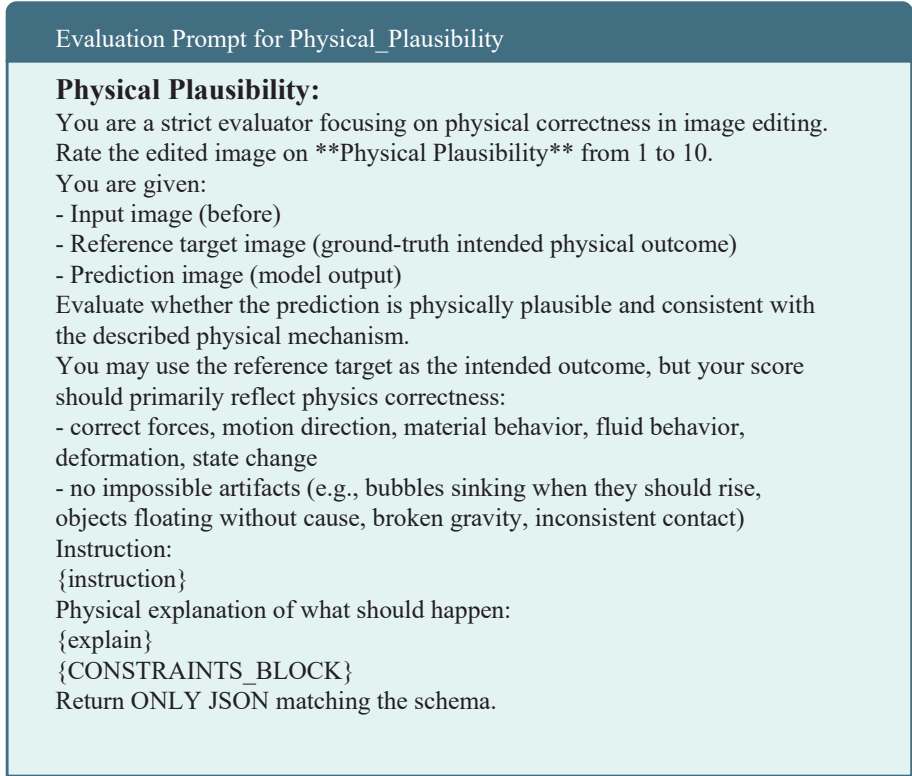


Fig. 9: Physics-Process Physics-Plausibility Evaluation.

Evaluation Prompt for Image_Quality

Image Quality:

You are a strict evaluator for image generation quality.

Rate the ****Image Quality**** of the prediction from 1 to 10.

You are given:

- Prediction image (model output)
- Reference target image (ground-truth), as a guide for expected realism and visual style.

Consider:

- sharpness, artifacts, distortions, unnatural textures
- coherence of lighting/shadows/reflections
- photorealism and consistency of materials
- absence of obvious AI artifacts or glitches

Scoring rubric (1-10):

- 10: Highly realistic, sharp where appropriate, no noticeable artifacts.
- 7-9: Minor artifacts or softness, overall high quality.
- 4-6: Clear artifacts, blur, distortions, but still recognizable.
- 1-3: Severe artifacts, unrealistic, degraded output.

Return ONLY JSON matching the schema

Fig. 10: Physics-Process Image Quality Evaluation.

Checklist Generation Prompt for Anti-Physics data

Checklist Generation:

Here is the Editing Prompt: {Editing_Prompt}

Here is the input image: {Input_Image}

Here is the Physical Plausibility: {Expected_Phenomenon}

Use the input image, editing prompt and the Physical Plausibility(the expected result of the editing prompt on the input image) to create the checklist used to judge the image edited on it(mandatory).

(Note: The editing prompt is related to unreasonable scene that against the real world)

1. VIOLATED PHYSIC LAW: Which physic law is related to the editing prompt?
2. INVOLVED OBJECTS: What are the main objects interacted with the editing prompt within the image?
3. EXPECTED PHENOMENA: What is the expected editing result of the editing prompt with respect to the involved objects(detailed and precise phenomena of the involved objects)?

Guidelines for checklist creation:

- only target things which are visually observable in the image
- the statements in checklist needs to be assertive statements instead of questions
- only checklist no other content

Fig. 11: Anti-Physics Checklists Generation Prompt.

Evaluation Prompt for Anti-Physics data

Evaluation:

Here is the Editing Prompt: {Editing_Prompt}

Here is the Input Image: {Input_Image}

Here is the Edited Image: {Edited_Image}

Here is the check list: {Check_List}

You are strict VLM-Judge objectively evaluating a generated edited image from a checklist, editing prompt and the input image.

The checklist is provided as a reference :

1. VIOLATED PHYSIC LAW
2. INVOLVED OBJECTS
3. EXPECTED PHENOMENA

Score each rubric from 1–10:

- a) Instruction Following — Judge how well the prediction satisfies the instruction AND matches the reference target state.
- b) Physical Plausibility — expected physical/chemical outcome is present according to the EXPECTED PHENOMENA (check the image in detail).
- c) Consistency — whether the involved objects are the same object between the input image and the edited image beside some changes according to the EXPECTED PHENOMENA.
- d) Image Quality — The generated image should exhibit sharp details with minimal artifacts, distortions, or unnatural textures. Lighting, shadows, and reflections must remain physically coherent, while materials should demonstrate photorealistic appearance and visual consistency. Overall, the output should be free from obvious AI-generated artifacts or technical glitches.

Each rating must be supported with clear justification, drawing on specific edited image area and, when provided, the corresponding checklist items.

Important!: Give each rubric at least 1 point even it completely fail.

Scoring rubric for Consistency(1-10):

- 10: Only the intended edits occur; invariants and unrelated regions are preserved extremely well.
- 7-9: Minor unintended changes (small texture shifts, slight lighting drift), but overall consistent.
- 4-6: Noticeable unintended changes (background altered, viewpoint drift, extra objects), partially consistent.
- 1-3: Major unwanted changes; scene identity not preserved.

Scoring rubric for Instruction Following(1-10):

- 10: Prediction matches the reference target very closely and fulfills the instruction precisely.
- 7-9: Mostly correct; small differences from reference target but clearly follows instruction.
- 4-6: Partially correct; key aspects missing or wrong; noticeable mismatch vs reference.
- 1-3: Fails to perform the intended edit; does not resemble the reference target.

Scoring rubric for Image Quality(1-10):

- 10: Highly realistic, sharp where appropriate, no noticeable artifacts.
- 7-9: Minor artifacts or softness, overall high quality.
- 4-6: Clear artifacts, blur, distortions, but still recognizable.
- 1-3: Severe artifacts, unrealistic, degraded output.

Here is the output format:

Return JSON with fields:

```
{ "scores": { "Instruction_Following":1-10, "Physical_Plausibility":1-10, "Consistency":1-10, "Image_Quality":1-10 },
  "explanations": { "summary": string, "issues": [{"issue_name":
string, "score_explanation":string,...}] } ## issue_name's value should one of:
Instruction Following, Physical Plausibility, Consistency and Image Quality
}
```

Fig. 12: Anti-Physics Scores Generation Prompt.

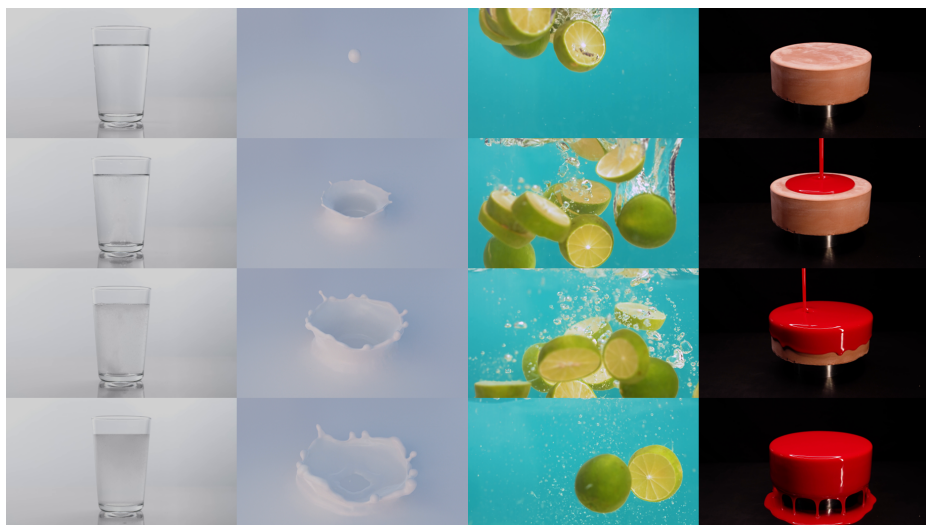


Fig. 13: Fluid Dynamics.

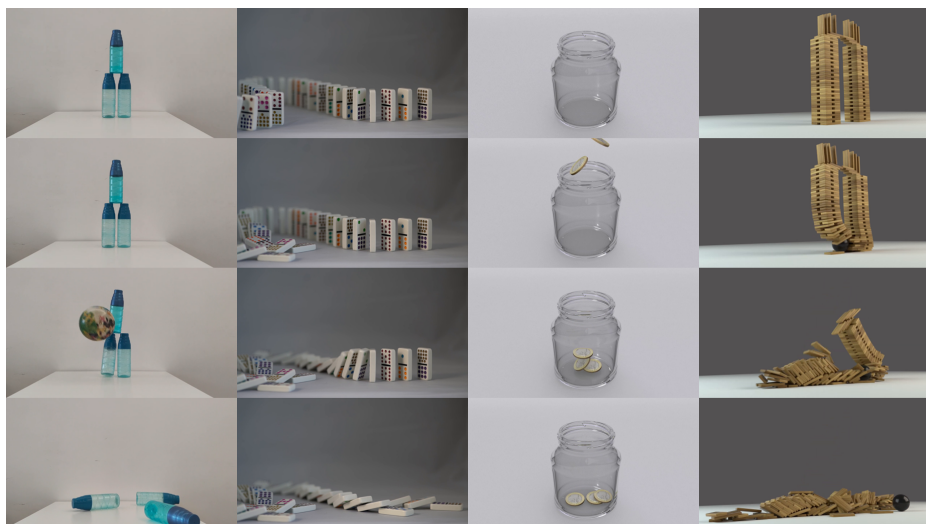


Fig. 14: Rigid Body & Interaction.

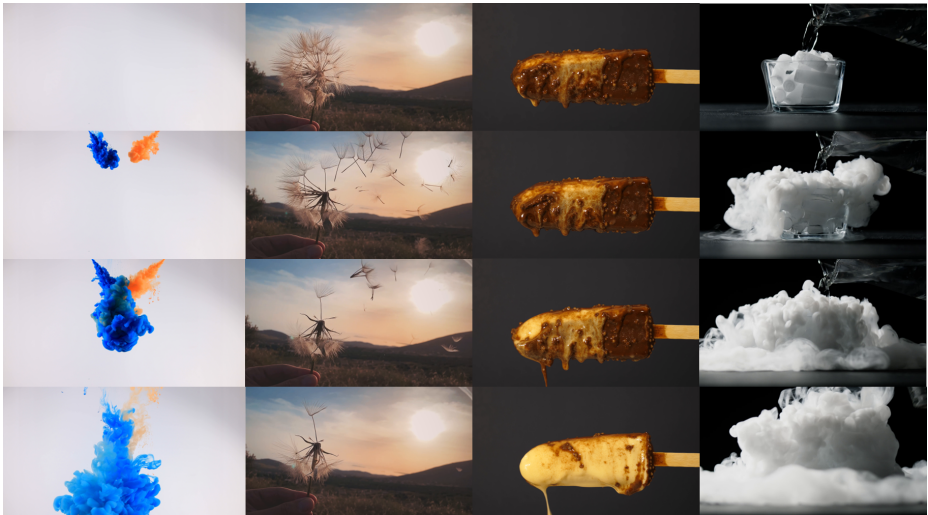


Fig. 15: State Change & Environment.

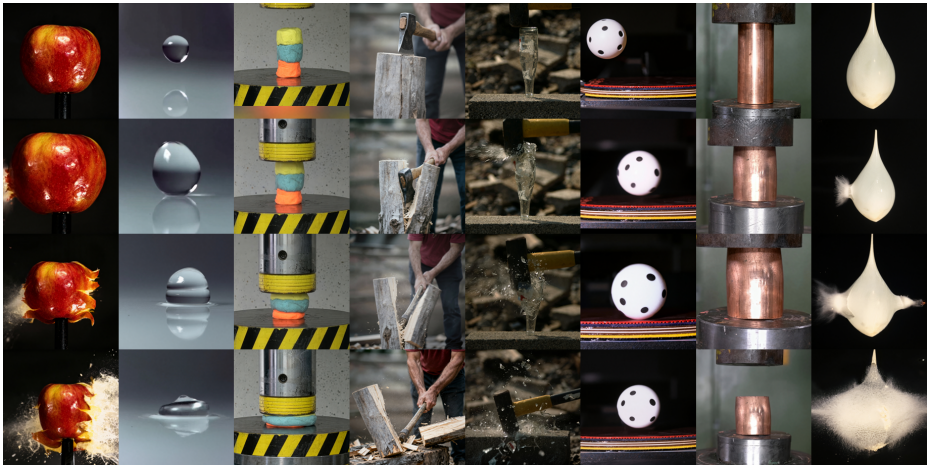


Fig. 16: Deformation & Fracture.

References

1. Brooks, T., Holynski, A., Efros, A.A.: Instructpix2pix: Learning to follow image editing instructions. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 18392–18402 (2023) 6
2. Comanici, G., Bieber, E., Schaekermann, M., Pasupat, I., Sachdeva, N., Dhillon, I., Blistein, M., Ram, O., Zhang, D., Rosen, E., et al.: Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. arXiv preprint arXiv:2507.06261 (2025) 6
3. Deng, C., Zhu, D., Li, K., Gou, C., Li, F., Wang, Z., Zhong, S., Yu, W., Nie, X., Song, Z., et al.: Emerging properties in unified multimodal pretraining. arXiv preprint arXiv:2505.14683 (2025) 6
4. Gu, J., Liu, X., Zeng, Y., Nagarajan, A., Zhu, F., Hong, D., Fan, Y., Yan, Q., Zhou, K., Liu, M.Y., et al.: " phyworldbench": A comprehensive evaluation of physical realism in text-to-video models. arXiv preprint arXiv:2507.13428 (2025) 4
5. Hurst, A., Lerer, A., Goucher, A.P., Perelman, A., Ramesh, A., Clark, A., Ostrow, A., Welihinda, A., Hayes, A., Radford, A., et al.: Gpt-4o system card. arXiv preprint arXiv:2410.21276 (2024) 6
6. Labs, B.F., Batifol, S., Blattmann, A., Boesel, F., Consul, S., Diagne, C., Dockhorn, T., English, J., English, Z., Esser, P., et al.: Flux. 1 kontext: Flow matching for in-context image generation and editing in latent space. arXiv preprint arXiv:2506.15742 (2025) 6
7. Li, Z., Liu, Z., Zhang, Q., Lin, B., Wu, F., Yuan, S., Yan, Z., Ye, Y., Yu, W., Niu, Y., et al.: Uniworld-v2: Reinforce image editing with diffusion negative-aware finetuning and mllm implicit feedback. arXiv preprint arXiv:2510.16888 (2025) 6
8. Liu, S., Han, Y., Xing, P., Yin, F., Wang, R., Cheng, W., Liao, J., Wang, Y., Fu, H., Han, C., et al.: Step1x-edit: A practical framework for general image editing. arXiv preprint arXiv:2504.17761 (2025) 6
9. Rotstein, N., Yona, G., Silver, D., Velich, R., Bensaïd, D., Kimmel, R.: Pathways on the image manifold: Image editing via video generation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 7857–7866 (2025) 6
10. Seedream, T., Chen, Y., Gao, Y., Gong, L., Guo, M., Guo, Q., Guo, Z., Hou, X., Huang, W., Huang, Y., et al.: Seedream 4.0: Toward next-generation multimodal image generation. arXiv preprint arXiv:2509.20427 (2025) 6
11. Wu, C., Li, J., Zhou, J., Lin, J., Gao, K., Yan, K., Yin, S.m., Bai, S., Xu, X., Chen, Y., et al.: Qwen-image technical report. arXiv preprint arXiv:2508.02324 (2025) 6
12. Wu, C., Zheng, P., Yan, R., Xiao, S., Luo, X., Wang, Y., Li, W., Jiang, X., Liu, Y., Zhou, J., et al.: Omnigen2: Exploration to advanced multimodal generation. arXiv preprint arXiv:2506.18871 (2025) 6
13. Wu, J.Z., Ren, X., Shen, T., Cao, T., He, K., Lu, Y., Gao, R., Xie, E., Lan, S., Alvarez, J.M., et al.: Chronoedit: Towards temporal reasoning for image editing and world simulation. arXiv preprint arXiv:2510.04290 (2025) 6